

Problem Set 2: Answers

Probability and Statistical Inference

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1. A continuous random variable X has density

$$f_X(x) = \begin{cases} c(1+x), & -1 \leq x < 0, \\ c(1-x/2), & 0 \leq x \leq 2, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a constant.

- (a) Find c and briefly explain why the resulting function is a valid density.
- (b) Derive the CDF $F_X(x)$ over the whole real line. Check that your expressions agree at $x = 0$ and that the CDF reaches one at the upper end of the support.
- (c) Sketch the density and the CDF. Label the support and the values at $x = -1, 0, 2$.
- (d) Find $\Pr(-1/2 < X \leq 1)$.
- (e) Calculate $\mathbb{E}[X]$ and $\text{Var}(X)$.
- (f) The density has height $f_X(0) = 2/3$. What is $\Pr(X = 0)$? Explain why these two quantities are different.

Solution. For part (a), non-negativity holds on both pieces. The total area is

$$c \int_{-1}^0 (1+x) dx + c \int_0^2 (1-x/2) dx = c \left(\frac{1}{2} + 1 \right),$$

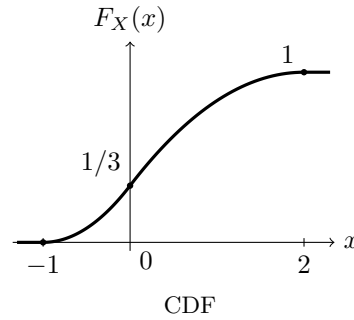
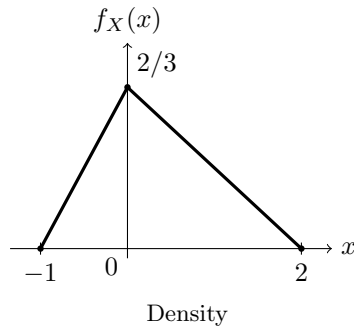
so $c = 2/3$. The function is therefore non-negative and integrates to one.

For part (b), integrating from the left endpoint gives

$$F_X(x) = \begin{cases} 0, & x < -1, \\ \frac{(x+1)^2}{3}, & -1 \leq x < 0, \\ \frac{1}{3} + \frac{2x}{3} - \frac{x^2}{6}, & 0 \leq x \leq 2, \\ 1, & x > 2. \end{cases}$$

Both middle expressions equal $1/3$ at zero, and the third expression equals one at $x = 2$.

For part (c), the density is a straight line from $(-1, 0)$ to $(0, 2/3)$ and then to $(2, 0)$; it is zero outside $[-1, 2]$. The CDF is flat at zero to the left of -1 . It then follows the two quadratic pieces in part (b), reaches one at $x = 2$, and remains flat thereafter.



For part (d), continuity of X gives

$$\Pr(-1/2 < X \leq 1) = F_X(1) - F_X(-1/2) = \frac{5}{6} - \frac{1}{12} = \frac{3}{4}.$$

For part (e),

$$\begin{aligned} \mathbb{E}[X] &= \frac{2}{3} \left[\int_{-1}^0 x(1+x) dx + \int_0^2 x(1-x/2) dx \right] \\ &= \frac{2}{3} \left(-\frac{1}{6} + \frac{2}{3} \right) = \frac{1}{3}, \end{aligned}$$

and

$$\begin{aligned} \mathbb{E}[X^2] &= \frac{2}{3} \left[\int_{-1}^0 x^2(1+x) dx + \int_0^2 x^2(1-x/2) dx \right] \\ &= \frac{2}{3} \left(\frac{1}{12} + \frac{2}{3} \right) = \frac{1}{2}. \end{aligned}$$

Consequently,

$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{2} - \frac{1}{9} = \frac{7}{18}.$$

For part (f), $\Pr(X = 0) = 0$. A density is probability per unit on the horizontal scale, not point probability. A single point has zero width, so it has zero probability under a continuous distribution even when the density there is positive.

2. Let $X \sim U[-2, 1]$ and define $Y = X^2$. The transformation is not one-to-one on the support of X : different values of X can produce the same value of Y .
 - (a) State the support of Y . For each possible value y , identify which of the roots $x = -\sqrt{y}$ and $x = \sqrt{y}$ lie in the support of X . Explain why the answer changes at $y = 1$.
 - (b) Use the CDF method to derive $F_Y(y)$ for all $y \in \mathbb{R}$.
 - (c) Hence derive the density $f_Y(y)$. Verify that it integrates to one.
 - (d) Calculate $\Pr(1/4 \leq Y \leq 9/4)$.
 - (e) Calculate $\mathbb{E}[Y]$ using your density and verify the result directly from $Y = X^2$.
 - (f) A student applies the lecture's density-transformation formula using only the inverse $x = \sqrt{y}$. Describe precisely which inverse branch is missed and for which values of y .

Solution. For part (a), the support is $[0, 4]$. When $0 < y \leq 1$, both $-\sqrt{y}$ and $\sqrt{y}$ belong to $[-2, 1]$. When $1 < y \leq 4$, only $-\sqrt{y}$ belongs to the support, because $\sqrt{y} > 1$. The change at one is caused by the shorter positive part of the support of X .

For part (b), $f_X(x) = 1/3$ on $[-2, 1]$. For $0 \leq y \leq 1$, the event $X^2 \leq y$ is the interval $[-\sqrt{y}, \sqrt{y}]$, of length $2\sqrt{y}$. For $1 < y \leq 4$, it is $[-\sqrt{y}, 1]$, of length $1 + \sqrt{y}$. Hence

$$F_Y(y) = \begin{cases} 0, & y < 0, \\ \frac{2\sqrt{y}}{3}, & 0 \leq y \leq 1, \\ \frac{1 + \sqrt{y}}{3}, & 1 < y \leq 4, \\ 1, & y > 4. \end{cases}$$

The two interior expressions both equal $2/3$ at $y = 1$.

For part (c), differentiation gives

$$f_Y(y) = \begin{cases} \frac{1}{3\sqrt{y}}, & 0 < y < 1, \\ \frac{1}{6\sqrt{y}}, & 1 < y < 4, \\ 0, & \text{otherwise.} \end{cases}$$

Its value at a single boundary point can be assigned arbitrarily. Its total area is

$$\int_0^1 \frac{1}{3\sqrt{y}} dy + \int_1^4 \frac{1}{6\sqrt{y}} dy = \frac{2}{3} + \frac{1}{3} = 1.$$

For part (d),

$$\Pr(1/4 \leq Y \leq 9/4) = F_Y(9/4) - F_Y(1/4) = \frac{5}{6} - \frac{1}{3} = \frac{1}{2}.$$

For part (e),

$$\begin{aligned} \mathbb{E}[Y] &= \int_0^1 \frac{y}{3\sqrt{y}} dy + \int_1^4 \frac{y}{6\sqrt{y}} dy \\ &= \frac{2}{9} + \frac{7}{9} = 1. \end{aligned}$$

Directly,

$$\mathbb{E}[X^2] = \frac{1}{3} \int_{-2}^1 x^2 dx = \frac{1}{3} \left[\frac{x^3}{3} \right]_{-2}^1 = 1,$$

as required.

For part (f), the omitted inverse branch is $x = -\sqrt{y}$. This branch is valid throughout $0 < y < 4$. For $0 < y < 1$, omitting it loses one of the two equally weighted inverse branches and halves the correct density. For $1 < y < 4$, it loses the only valid inverse branch, since the positive root is then outside the support of X .

3. The joint distribution of discrete random variables $X \in \{0, 1\}$ and $Y \in \{0, 1, 2\}$ is shown below.

	$X = 0$	$X = 1$	$\Pr(Y = y)$
$Y = 0$	$1/4$	0	
$Y = 1$	$1/4$	$1/4$	
$Y = 2$	0	$1/4$	
$\Pr(X = x)$			1

- (a) Complete the marginal probabilities in the table.

- (b) Find the conditional distribution of Y given $X = 0$ and the conditional distribution of Y given $X = 1$.
- (c) For each $x \in \{0, 1\}$, calculate $\mathbb{E}[Y \mid X = x]$, $\mathbb{E}[Y^2 \mid X = x]$, and $\text{Var}(Y \mid X = x)$.
- (d) Calculate $\mathbb{E}[Y]$ directly from its marginal distribution and verify the law of iterated expectations, $\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}(Y \mid X)]$.
- (e) Calculate $\text{Var}(Y)$ directly and then verify the law of total variance,

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid X)] + \text{Var}(\mathbb{E}[Y \mid X]).$$

Interpret the two terms as within- X and between- X variation.

- (f) Are X and Y independent? Are they uncorrelated? Justify both answers.

Solution. For part (a), the completed table is

	$X = 0$	$X = 1$	$\Pr(Y = y)$
$Y = 0$	1/4	0	1/4
$Y = 1$	1/4	1/4	1/2
$Y = 2$	0	1/4	1/4
$\Pr(X = x)$	1/2	1/2	1.

For part (b), dividing each column by its marginal probability 1/2 gives

	$Y = 0$	$Y = 1$	$Y = 2$
$\Pr(Y = y \mid X = 0)$	1/2	1/2	0
$\Pr(Y = y \mid X = 1)$	0	1/2	1/2.

For part (c),

x	$\mathbb{E}[Y \mid X = x]$	$\mathbb{E}[Y^2 \mid X = x]$	$\text{Var}(Y \mid X = x)$
0	1/2	1/2	1/4
1	3/2	5/2	1/4.

For example, at $X = 1$, $\mathbb{E}[Y^2 \mid X = 1] = (1^2 + 2^2)/2 = 5/2$, so the variance is $5/2 - (3/2)^2 = 1/4$.

For part (d), the marginal distribution gives

$$\mathbb{E}[Y] = 0 \left(\frac{1}{4} \right) + 1 \left(\frac{1}{2} \right) + 2 \left(\frac{1}{4} \right) = 1.$$

The iterated calculation gives the same result:

$$\mathbb{E}[\mathbb{E}(Y \mid X)] = \frac{1}{2} \left(\frac{1}{2} \right) + \frac{1}{2} \left(\frac{3}{2} \right) = 1.$$

For part (e), $\mathbb{E}[Y^2] = 1/2 + 4/4 = 3/2$, and hence

$$\text{Var}(Y) = \frac{3}{2} - 1^2 = \frac{1}{2}.$$

The average within- X variance is

$$\mathbb{E}[\text{Var}(Y \mid X)] = \frac{1}{2} \left(\frac{1}{4} \right) + \frac{1}{2} \left(\frac{1}{4} \right) = \frac{1}{4}.$$

The conditional mean is equally likely to be 1/2 or 3/2, and its overall mean is one, so

$$\text{Var}(\mathbb{E}[Y \mid X]) = \frac{1}{2}(1/2 - 1)^2 + \frac{1}{2}(3/2 - 1)^2 = \frac{1}{4}.$$

Thus $1/2 = 1/4 + 1/4$. Half the total variance is average variation within the two X groups and half is variation between their conditional means.

For part (f), the variables are not independent. For example,

$$\Pr(X = 0, Y = 2) = 0 \neq \Pr(X = 0)\Pr(Y = 2) = \frac{1}{8}.$$

Also,

$$\mathbb{E}[X] = \frac{1}{2}, \quad \mathbb{E}[Y] = 1, \quad \mathbb{E}[XY] = \frac{1}{4} + \frac{2}{4} = \frac{3}{4},$$

so

$$\text{Cov}(X, Y) = \frac{3}{4} - \frac{1}{2}(1) = \frac{1}{4} \neq 0.$$

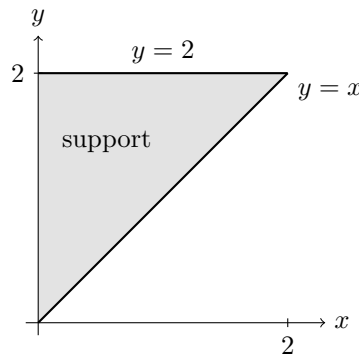
They are therefore correlated.

4. Suppose that (X, Y) has joint density

$$f_{X,Y}(x, y) = \begin{cases} cy, & 0 < x < y < 2, \\ 0, & \text{otherwise.} \end{cases}$$

- Draw the support of (X, Y) and find c .
- Derive the marginal densities $f_X(x)$ and $f_Y(y)$, stating their supports.
- Derive $f_{X|Y}(x | y)$ and $\mathbb{E}[X | Y = y]$ for $0 < y < 2$. Give an intuitive description of this conditional distribution.
- On a second sketch of the support, shade the event $X + Y \leq 2$. Explain why an integral over this event must be split at $y = 1$.
- Calculate $\Pr(X + Y \leq 2)$.
- Are X and Y independent? Give one argument based on either the support or the conditional density.

Solution. For part (a), the support is the triangle above the line $y = x$, to the right of $x = 0$, and below $y = 2$. The boundary lines are not included, but this does not affect any of the integrals.



Normalization gives

$$1 = \int_0^2 \int_0^y cy \, dx \, dy = c \int_0^2 y^2 \, dy = \frac{8c}{3},$$

so $c = 3/8$.

For part (b), for $0 < x < 2$,

$$f_X(x) = \int_x^2 \frac{3y}{8} \, dy = \frac{3}{16}(4 - x^2),$$

and $f_X(x) = 0$ otherwise. For $0 < y < 2$,

$$f_Y(y) = \int_0^y \frac{3y}{8} dx = \frac{3y^2}{8},$$

and $f_Y(y) = 0$ otherwise.

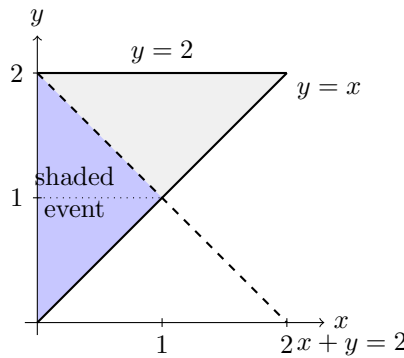
For part (c),

$$f_{X|Y}(x | y) = \frac{(3/8)y}{(3/8)y^2} = \frac{1}{y}, \quad 0 < x < y.$$

Thus, conditional on $Y = y$, X is uniform on $(0, y)$, and

$$\mathbb{E}[X | Y = y] = \int_0^y x \frac{1}{y} dx = \frac{y}{2}.$$

For part (d), the shaded triangle below is the part of the support that also satisfies $x + y \leq 2$.



At a fixed y , the original support imposes $0 < x < y$, while the event imposes $0 < x \leq 2 - y$. Hence the upper limit is $\min\{y, 2 - y\}$. The two possible upper limits are equal at $y = 1$, which is why the integral must be split there.

For part (e),

$$\begin{aligned} \Pr(X + Y \leq 2) &= \int_0^1 \int_0^y \frac{3y}{8} dx dy + \int_1^2 \int_0^{2-y} \frac{3y}{8} dx dy \\ &= \frac{1}{8} + \frac{1}{4} = \frac{3}{8}. \end{aligned}$$

For part (f), the variables are not independent. The possible values of X depend on Y : for example, when $Y = y$, X is restricted to $(0, y)$. Equivalently, $f_{X|Y}(x | y) = 1/y$ and its support depend on y , so it cannot equal one fixed marginal density $f_X(x)$ for every y .

5. This question studies the chain

$$X \perp\!\!\!\perp Y \implies Y \text{ is mean independent of } X \implies \text{Cov}(X, Y) = 0.$$

Assume throughout that the expectations used below exist.

- Prove both implications in the displayed chain.
- Let X be equally likely to be 0 or 1. Conditional on $X = 0$, let Y be equally likely to be -2 or 2 ; conditional on $X = 1$, let Y be equally likely to be -1 or 1 . Show that Y is mean independent of X , but that X and Y are not independent.

- (c) Let $X \sim U[-\pi, \pi]$ and set $Y = \cos X$. Show that X and Y are uncorrelated but that Y is not mean independent of X .
- (d) Explain how parts (b) and (c) establish that neither reverse implication is valid in general.

Solution. For part (a), independence means that the conditional distribution of Y given $X = x$ equals the marginal distribution of Y . Averaging Y under these identical distributions gives

$$\mathbb{E}[Y \mid X = x] = \mathbb{E}[Y]$$

for values of x where the conditional distribution is defined, which is mean independence. If Y is mean independent of X , the law of iterated expectations gives

$$\mathbb{E}[XY] = \mathbb{E}[X\mathbb{E}(Y \mid X)] = \mathbb{E}[X]\mathbb{E}[Y].$$

Therefore $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0$.

For part (b),

$$\mathbb{E}[Y \mid X = 0] = \frac{-2 + 2}{2} = 0, \quad \mathbb{E}[Y \mid X = 1] = \frac{-1 + 1}{2} = 0.$$

It follows by iterated expectations that $\mathbb{E}[Y] = 0$, so Y is mean independent of X . Nevertheless,

$$\Pr(Y = 2 \mid X = 0) = \frac{1}{2} \quad \text{whereas} \quad \Pr(Y = 2) = \frac{1}{4}.$$

The conditional and marginal distributions differ, so X and Y are not independent.

For part (c), symmetry gives

$$\mathbb{E}[X] = 0, \quad \mathbb{E}[Y] = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos x \, dx = 0.$$

Also $x \cos x$ is an odd function, so

$$\mathbb{E}[XY] = \frac{1}{2\pi} \int_{-\pi}^{\pi} x \cos x \, dx = 0.$$

Thus $\text{Cov}(X, Y) = 0$. But Y is completely determined by X , so

$$\mathbb{E}[Y \mid X] = \cos X,$$

which is not the constant $\mathbb{E}[Y] = 0$. Mean independence fails.

For part (d), part (b) satisfies mean independence without full independence, disproving the first reverse implication. Part (c) satisfies zero covariance without mean independence, disproving the second reverse implication. These examples do not contradict the forward implications proved in part (a).

6. A university recruits 200 volunteers for a study of an online revision workshop. Within this group, 100 students are randomly assigned to the workshop ($D = 1$) and 100 to the control group ($D = 0$). Let Y be the end-of-term score. Write the outcome in terms of its two group means:

$$Y = \alpha + \beta D + \varepsilon, \quad \mathbb{E}[\varepsilon \mid D] = 0.$$

The realized sample means are $\bar{Y}_1 = 76$ and $\bar{Y}_0 = 70$.

- (a) Show that $\alpha = \mathbb{E}[Y \mid D = 0]$ and $\beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0]$.
- (b) Estimate each population group mean by its sample group mean. Use part (a) to obtain $\hat{\alpha}$ and $\hat{\beta}$.

- (c) Using potential-outcome notation, state the random-assignment condition that allows a causal interpretation of the difference in means. For which population does the experiment identify an average causal effect directly?
- (d) In this description of the group means, interpret $\mathbb{E}[\varepsilon \mid D] = 0$ and show that it implies $\mathbb{E}[D\varepsilon] = 0$.
- (e) Does random assignment make the 200 volunteers representative of all university students? Explain the different roles of random assignment and random sampling.
- (f) Now suppose that the university took a random sample of students, but each student chose whether to attend the workshop. What would random sampling achieve? What additional problem would prevent an automatic causal interpretation of $\hat{\beta}$?

Solution. For part (a), take conditional expectations of the displayed equation. When $D = 0$,

$$\mathbb{E}[Y \mid D = 0] = \alpha + \mathbb{E}[\varepsilon \mid D = 0] = \alpha.$$

When $D = 1$,

$$\mathbb{E}[Y \mid D = 1] = \alpha + \beta + \mathbb{E}[\varepsilon \mid D = 1] = \alpha + \beta.$$

Subtracting the first equality from the second gives the stated expression for β .

For part (b), replace the two population conditional means in part (a) by their sample counterparts:

$$\hat{\alpha} = \bar{Y}_0 = 70, \quad \hat{\beta} = \bar{Y}_1 - \bar{Y}_0 = 76 - 70 = 6.$$

For part (c), writing $Y(d)$ for the score that would be observed under assignment d , the key condition is

$$D \perp\!\!\!\perp (Y(0), Y(1))$$

within the randomized study, together with positive probabilities of both assignments. Under random assignment, the treated and control means provide comparable averages of the two potential outcomes. The experiment directly identifies the average treatment effect for the recruited study participants. Applying that effect to other students requires random sampling or additional assumptions about whether their average treatment effect is the same.

For part (d), zero conditional mean says that the error ε has mean zero in each assignment group. The two conditional means are α and $\alpha + \beta$. Defining α and β from those means makes the remaining error average to zero within each group. By itself, it does not say that the groups have comparable potential outcomes or that β is causal. Moreover,

$$\mathbb{E}[D\varepsilon] = \mathbb{E}[\mathbb{E}(D\varepsilon \mid D)] = \mathbb{E}[D\mathbb{E}(\varepsilon \mid D)] = 0.$$

Thus the expected product of the treatment indicator and the error is zero.

For part (e), no. Random assignment determines which recruited participant receives treatment. It makes the treatment and control groups comparable within the study. Random sampling determines who enters the data and supports generalization to a target population. Here, students volunteered rather than being sampled at random, so they may differ systematically from all university students.

For part (f), a random sample would make the observed students representative of the target population under the sampling design. This supports generalization of descriptive quantities. It would not make students who choose treatment comparable with students who do not. The two groups might have different potential scores, motivation, or prior attainment, so D could be related to $(Y(0), Y(1))$. This selection problem prevents an automatic causal interpretation of $\hat{\beta}$.

In the regression with an intercept and a binary regressor, the error can still be defined as

$$\varepsilon = Y - \mathbb{E}[Y \mid D],$$

so $\mathbb{E}[\varepsilon \mid D] = 0$ continues to hold by construction. This equality of conditional means does not establish that the groups have comparable potential outcomes. The coefficient can still reflect selection bias.